

Automated Network Request Management in ServiceNow

Technology Stack & Architecture Overview

Date	09 September 2026
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Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4

1. Architecture Summary

The **Automated Network Request Management in ServiceNow** project is developed using the native capabilities of the ServiceNow platform. The solution provides a structured process for submitting and processing network requests.

A requester submits a Network Request through the ServiceNow Service Portal. The submitted request starts the configured workflow automation, which processes the request information and creates a corresponding record in the Network Database. The request is then sent for approval. Based on the approval decision, the workflow updates the record and sends the required notification.

The solution does not depend on external applications, APIs, machine-learning models, or third-party services.

2. Layer-by-Layer Breakdown

Layer	Purpose	Technology Used
User / Client Layer	Provides the interface through which the requester accesses and submits the network request.	ServiceNow Service Portal
Service Catalog Layer	Provides the Network Request catalog item and collects the information required from the requester.	ServiceNow Service Catalog
Form Validation Layer	Handles mandatory fields and controls field behaviour based on the values entered by the requester.	Catalog Configuration, UI Policies / Client-side configuration
Automation Layer	Processes the submitted request automatically, creates the database record, sends the approval request, handles the decision, and updates the record.	ServiceNow Workflow / Flow Designer
Data Layer	Stores the submitted network request information and maintains the request status and approval information.	Custom Network Database Table
Approval Layer	Sends the created network request for review and captures the approver's decision.	ServiceNow Approval
Decision Layer	Determines the next step based on whether the request is approved or rejected.	If/Else workflow logic
Notification Layer	Sends email notifications to the configured users or stakeholders during the request process.	ServiceNow Email / Workflow Notification
Security Layer	Controls access to the network request records based on authorised users and roles.	ServiceNow Roles and ACLs
Runtime Environment	Provides the environment in which the application is configured, tested, and demonstrated.	ServiceNow Personal Developer Instance (PDI)

External Integration Layer	No external application or third-party system is required for the implemented solution.	Not applicable
AI / Intelligence Layer	No AI, machine-learning, or predictive functionality is included in the current project.	Not applicable

3. Architecture Diagram

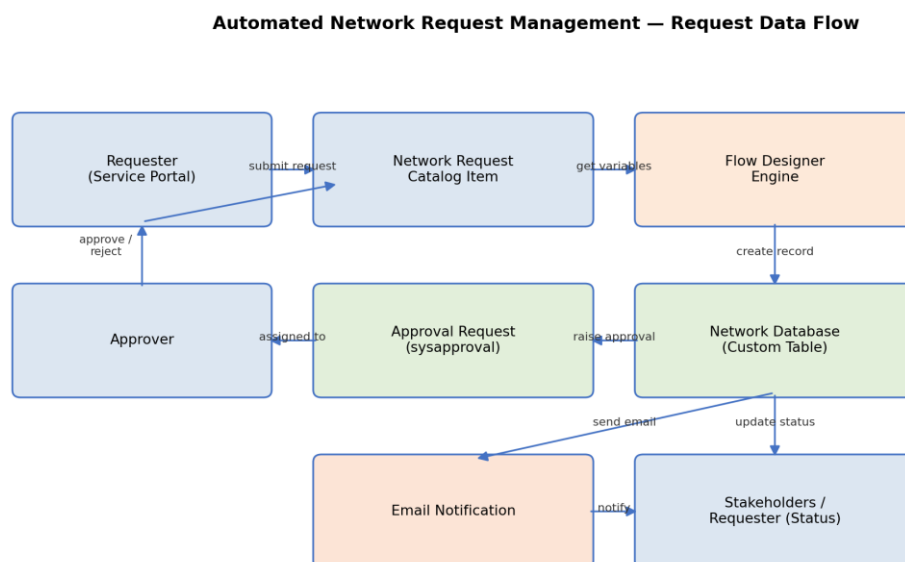


Figure 1: Request flow across the ServiceNow components listed above.

4. Component Interaction

Service Portal

The Service Portal acts as the entry point for the requester. The requester accesses the Network Request catalog item and provides the required information.

Network Request Catalog Item

The catalog item collects the network request information entered by the requester. Mandatory fields and configured form behaviour help ensure that the required information is provided before submission.

Workflow Automation

After submission, the configured ServiceNow workflow automation processes the request. It retrieves the submitted information, creates the corresponding Network Database record, initiates the approval process, evaluates the approval result, and performs the required record updates.

Network Database

The custom Network Database table stores the request information in a structured ServiceNow record. It also maintains information related to the request status and approval outcome.

Approval

The approval step sends the network request to the configured approver. The approver reviews the request and provides an approval or rejection decision.

If/Else Decision

The workflow evaluates the approval result using conditional logic. The appropriate branch is executed depending on whether the request is approved or rejected.

Record Update

After the approval decision, the Network Database record is automatically updated so that the latest status is available for tracking.

Notification

The configured notification step sends an email to the relevant recipients, keeping them informed about the request process or outcome.

5. Notes on Scope

- The solution is implemented entirely within the ServiceNow platform.
- No external REST or SOAP API integration is used.
- No third-party application or external database is connected to the project.
- No AI, machine-learning, or predictive component is included.
- The workflow is designed specifically to automate the network request lifecycle.
- Development, configuration, testing, and demonstration are performed in a ServiceNow Personal Developer Instance.
- The architecture focuses on request submission, automated processing, database record creation, approval, decision handling, status update, and notification.